

Experiment Instruction

Software Engineering Economics, Experiment 1

Seeking Demand Curve

1. Background

This experiment will simulate a perfectly competitive market, the Goods transacted in this market is book. The goal of this experiment is to find the demand & supply curve in book market by auction. Teacher plays the role of supplier of book. Students will participate in book auctions in the market as demanders or suppliers. By quoting, students will understand the behavioral characteristics of the demand side and supply side in a perfectly competitive market.

After completing this experiment, students will achieve the following goals

1.1 Familiar with the market auction mechanism.

1.2 Master the ability to record experimental data, organize data, and obtain market transaction information.

1.3 Able to calculate the best price of book transaction based on the maximum benefit of the supplier.

1.4 Able to draw the demand curves based on all experimental data.

This experiment is an in-class verification experiment project, with One teaching hour for experiment process in classroom and another One hour for completing the experimental report out-of-class.

2. Experimental Principle

Seek how to maximize the profit of supplier according to the behavior of students participating in the experiment. Assuming that the trading product (a specific topic books) owned by teacher and he want to sell to the students, the students do not have to bear any cost in the trading process.

Assume the transaction cost of this experiment is Zero, that is to say, the profit earned by the teacher will be equal to the specific transaction price multiplied by the number of books that may be sold to the student at this specific price.

3. Experimental Rules

3.1 Teacher announces the rules of auction and experimental procedures to students, guides students to become familiar with the auction transaction, and asks students not to collude with each other during the auction process (otherwise, the student will be disqualified from participating in this experiment).

3.2 Using the auction method, students (Demanders) can purchase trading product (a specific book) at any price. But in teacher side (Supplier), too high prices result in a

small number of books being sold, leading to low profits for supplier; The low price may result in a large number of books being sold, but due to the low selling price of individual books, the profit for the teacher will not be significant. Therefore, teacher should choose auction method to sell the books, observes consumer's (student's) behavior in the auction process, finds a suitable price to maximize the profit at this price level.

In this experiment, student quotations should be reasonable, usually not exceeding 500 yuan, and should not be lower than 0 yuan.

3.3 (Extension) Vickery auction. Vickery auction is also known as Second price sealed-bid auction. An auctioneer makes a bid without knowing the bids of the others (sealed bids). The person with the highest bid gets the right to buy, but what he actually pays is, the second price, that is the bid second to the highest bid. This rule will incentivize each bidder to give their true willingness to purchase, provided it does not exceed the demander's maximum willingness to pay.

4. Experiment Steps

Step 1: The teacher displays the trading items. In this experiment, he needs to display a specific book which is ready to participate in the auction (can be a single book or multiple copies of the same book). Teachers need to explain that the price of this book is only a nominal price, not the real price in market transaction. And students do not need to pay any costs in the auction.

Step 2: The teacher explains the experimental principle and rules. If there is enough time, he can also introduce the Vickrey auction method.

Step 3: Teacher tells the students participating in this experiment, acted as market demanders, that they can quote based on their true willingness to get this book. Teacher also needs to tell the students participating in the experiment that they are not allowed to collude with quotation information.

Step 4: Students fill in the auction quotation form based on their own preferences for the book. They can use a piece of paper for quotation (or fill in the quotation online). After all of the quotation are filled in, all of quotation forms need to be sealed and handed over to the teacher (or upload quotations online).

Step 5: Teacher invites the student representatives who participated in the experiment, compile the quotation list, check the quotations, and then publicly count the quotations.

Step 6: When the student representatives are counting quotations, teacher needs to publicly enter the quotation information one by one into a table which is processing the quotation data in real time. During the quotation counting process, if an extreme quotation (number) is found, such as 0 yuan (and below) or 500 yuan (and above), this quotation will be regarded as an invalid quotation, and the quotation information will not be used as valid information in this experiment.

Step 7: After the voting for all quotations is completed, the teacher organizes the data into transaction information on the demand side, generates demand curve, and showing this curve to students publicly.

Step 8: Teacher needs to explain to the students participating in the experiment the operation process of how to organize the experimental data into auction transaction information, and how to generate the demand curve of this auction and tell students the meaning of the demand curve.

Step 9: Based on the result of transaction prices obtained from the auction, teacher calculates the maximum profit, find the suitable price. At this price, the supplier (teacher) will obtain the maximum potential profit.

The maximum profit is equal to the suitable price multiplied by the transaction volume corresponding to this price.

Step 10: Teacher found the student (the lucky person) who provided the appropriate quote and commended her/him. At the same time, teachers can inspire students to discuss how demand curve may change if the transaction prices changed.

Step 11 (Extension): Find the intersection point of the supply curve and the demand curve obtained in this experiment. The price corresponding to this point is the transaction price of the supply side and demand side participating in the experiment when the market is in equilibrium.

5. Experiment Report

Students must submit a hand-written experimental report (PDF format) on Canvas platform in one week after completing this experiment.

The experimental report must include the following 4 required contents:

- Experimental background.
- Experimental content and steps.
- Experimental results (Demand Curve) and your bid price.
- Your learning experiences in this Experiment.

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